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↓
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⓪ $V$ : vector space
$v_0, v_1, \dots, v_n \in V$ : vectors.
$\{v_i\}_{i=0}^{n} \xleftrightarrow[\text{identify}]{} E=\{0,1,\dots,n\}$
$\mathcal{L} := \{\mathrm{Span}(S)\cap E \,;\, S\subset V\}$
$(E,\mathcal{L})$ : matroid.

# Poincaré duality of matroids via star subdivisions (survey)

§ Intro

$n\in\mathbb{Z}_{\geq 0}$, $E:=\{0,1,\dots,n\}$, $\mathcal{L}\subset 2^{E}$

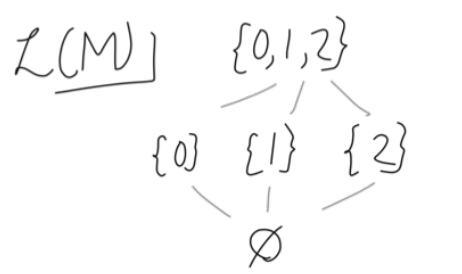
Def
$M:=(E,\mathcal{L})$ is called a loopless matroid
$\overset{\text{def}}{\iff}$ ① $\emptyset, E\in\mathcal{L}$
② $F,G\in\mathcal{L} \Rightarrow F\cap G\in\mathcal{L}$
③ $F\in\mathcal{L}$, $e\in E\setminus F$
$\Rightarrow \exists G\in\mathcal{L}$ s.t. $\begin{cases} G \underset{\text{cover}}{\supset} F, \\ e\in G \end{cases}$
$\nexists H\in\mathcal{L}$ s.t. $F\subsetneq H\subsetneq G$

- subsets belonging $\mathcal{L}$ are called flats of $M$
- $\mathcal{L}(M):=\mathcal{L}$
  the set of flats of $M$
- $(\mathcal{L}(M), \subset)$ is a lattice
- all maximal chain of flats have the same length, called the rank of $M$

Example
❶ $\mathcal{L}(M):=2^{E}$
($M$ is called a Boolean matroid) $B_n:=M$
- $\mathrm{rank}(B_n)=|E|=n+1$

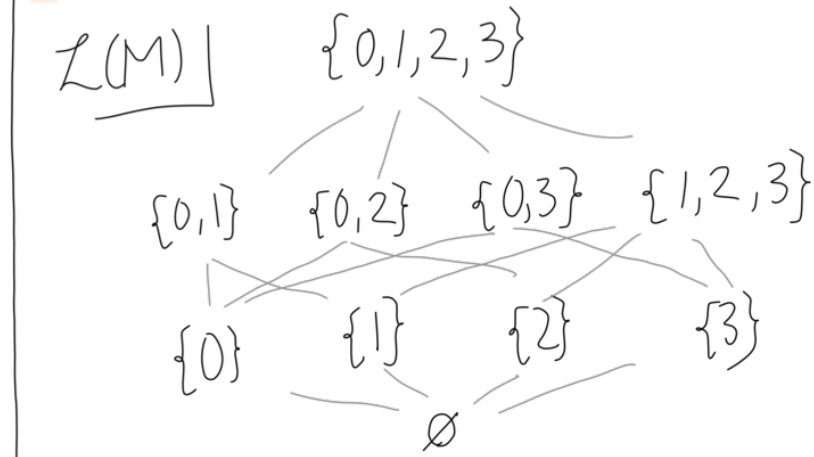
$\emptyset\subsetneq\{0\}\subsetneq\{0,1\}\subsetneq\cdots\subsetneq\{0,1,\dots,n\}=E$

❷ $n=2$, $E=\{0,1,2\}$

$\mathcal{L}(M)$


- $\mathrm{rank}(M)=2$

❸ $n=3$, $E=\{0,1,2,3\}$

$\mathcal{L}(M)$


- $\mathrm{rank}(M)=3$

- Matroids induces a polyhedral fan

$\bigoplus_{i\in E}\mathbb{Z}e_i$ : the free abelian grp on $E$.
rank $=n+1$

For $S\subset E$, $e_S:=\sum_{i\in S}e_i$

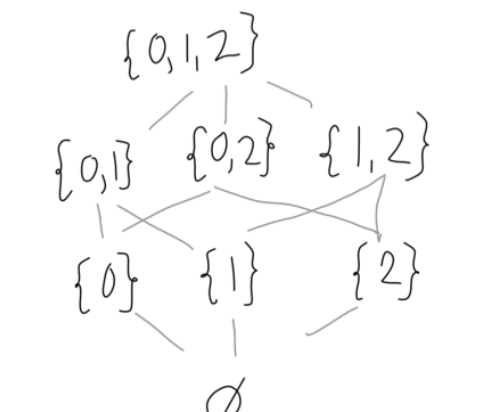
$N_E := \dfrac{\bigoplus_{i\in E}\mathbb{Z}e_i}{\mathbb{Z}e_E}$ ← rank $n$

$N_{E,\mathbb{R}} := N_E \otimes_{\mathbb{Z}} \mathbb{R}$

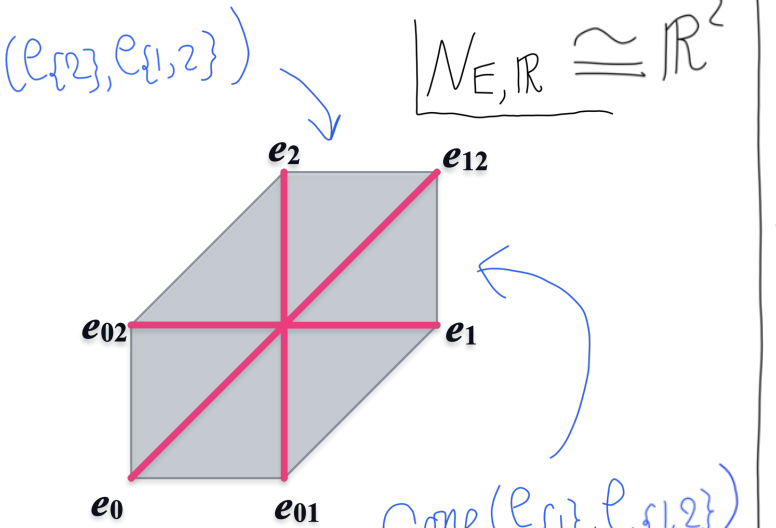
Def $\mathcal{P}(M):=\mathcal{L}(M)\setminus\{\emptyset, E\}$
$\Sigma_M := \{\mathrm{Cone}(e_F ; F\in\mathcal{F}) ; \mathcal{F} \underset{\text{flag}}{\subset} \mathcal{P}(M)\}$
$\cap$
$N_{E,\mathbb{R}}$
(being totally ordered subset w.r.t. $\subset$)

- $\Sigma_M$ is called the Bergman fan of $M$
- $\Sigma_M$ is a unimodular fan of pure dim. $=\mathrm{rk}(M)-1$
  corresp. toric var is smooth

Exa.
❶ $n=2$, $B_2$: Boolean ($\mathcal{L}(B_2)=2^{\{0,1,2\}}$)

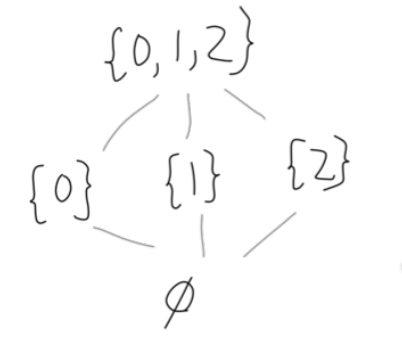
$\mathcal{L}(B_2)$


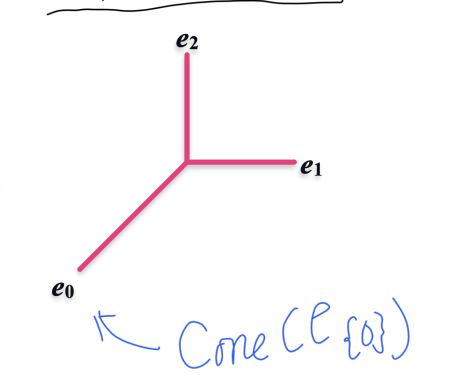
$\mathrm{rank}(B_2)=3$

$N_{E,\mathbb{R}} \cong \mathbb{R}^2$


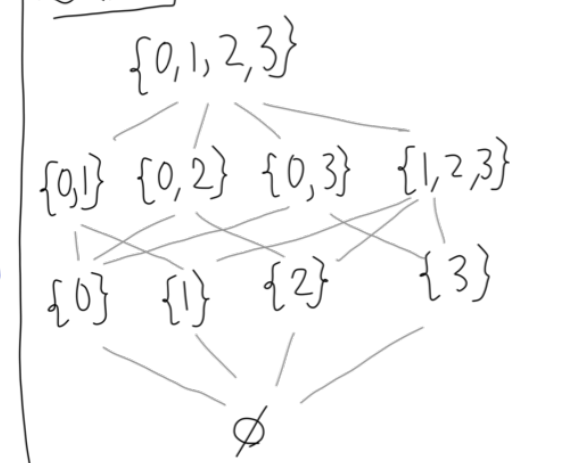
$e_0 = -e_1-e_2$ $\dim\Sigma_{B_2}=2$
$e_0+e_1+e_2=0$

❷ $n=2$

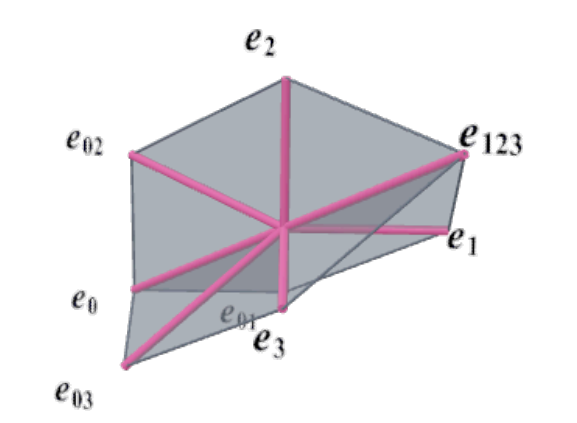
$\mathcal{L}(M)$


$N_{E,\mathbb{R}}\cong\mathbb{R}^2$


❸ $n=3$

$\mathcal{L}(M)$


$\mathrm{rank}(M)=3$

$N_{E,\mathbb{R}}\cong\mathbb{R}^3$


$\dim\Sigma_M=2$

$X(\Sigma_M)$ := the toric variety $/\mathbb{C}$ corresponding to $\Sigma_M\subset N_{E,\mathbb{R}}$
of dim $n$, smooth, quasi-projective but non-proper in general

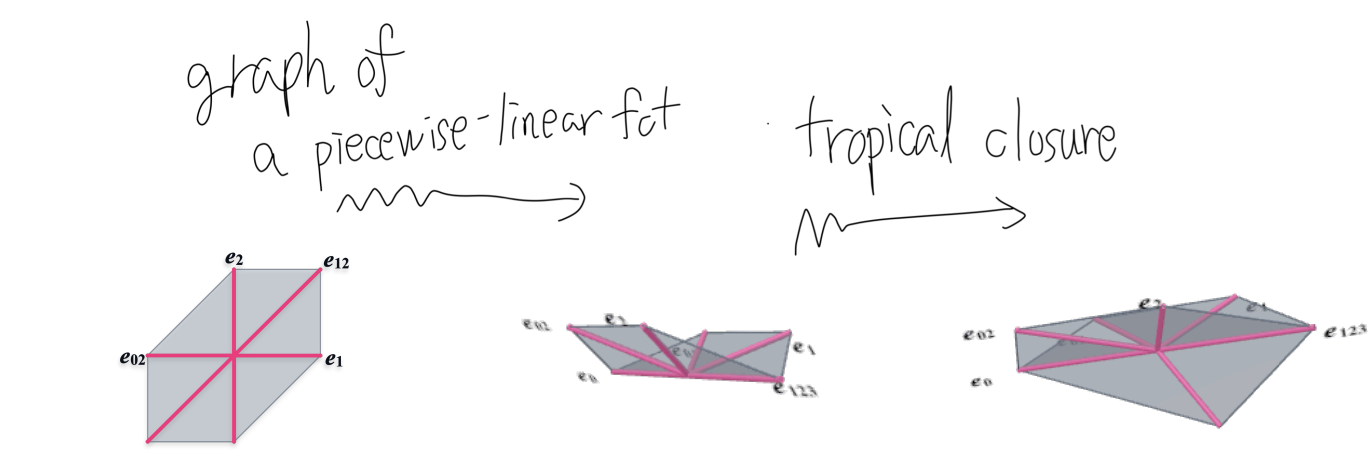
Def
$A^{\bullet}(M) := A^{\bullet}(X(\Sigma_M)) \cong \dfrac{\mathbb{Z}[x_F]_{F\in\mathcal{P}(M)}}{(\text{some relations})}$
the Chow ring of $X(\Sigma_M)$
called the Chow ring of $M$

We can study Char poly of $M$ (chrom. poly of a graph) via $A^{\bullet}(M)$

Thm (Adiprasito-Huh-Katz '18)
Let $M$ be a loopless matroid on $E$, rank $=r+1$
$A^{\bullet}(M)$ satisfies the Kähler package :
- Poincaré duality of dim $r$ ← We talk today
- hard Lefschetz property
- Hodge-Riemann relation

We do not talk today

- $q>r \Rightarrow A^{q}(M)=0$
- $A^{r}(M)\cong\mathbb{Z}$
- if $q\leq r$,
  $A^{q}(M)\times A^{r-q}(M)\longrightarrow A^{r}(M)\cong\mathbb{Z}$
  $(x,y)\longmapsto xy$
  is a perfect pairing



§ Proof of Thm

History: AHK'18 $\rightsquigarrow$ Amini-Piquerez '20 $\rightsquigarrow$ Hsiao '25
another method — simplify

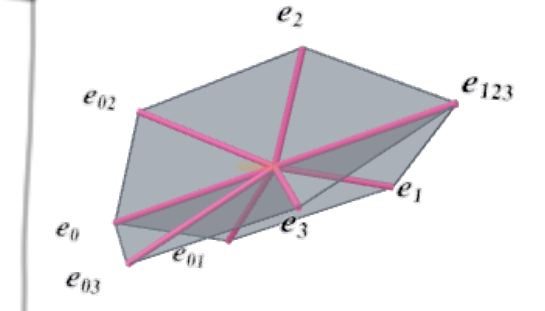
- AHK'18 method : induction on $n$ & $r$
  $n=|E|-1$, $r=\mathrm{rank}(M)-1$
  matroidal flip
- Amini-Piqurez '20 method : induction on $n$
  Ⓘ tropical modification (Deletion-Contraction of Matroids)
  Ⓘ Weak factorization thm for smooth toric var (Blowup center?)
  Ⓘ Keel's lemma for PD
- Hsiao '25 method :
  Construct an explicit sequence of Blowups,
  in particular, explicit Blowup centers

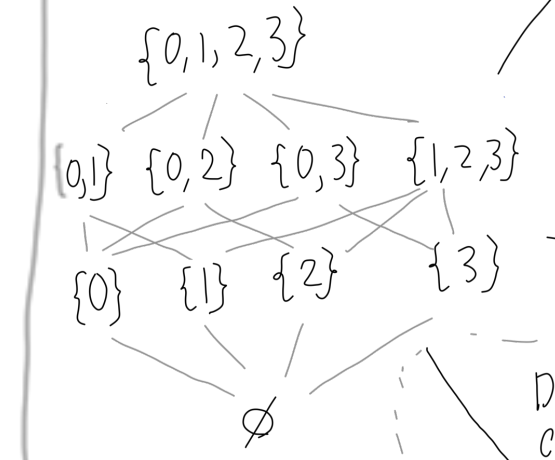
Below, we may assume that
$M$ is simple, i.e., $\forall a\in E$, $\{a\}\in\mathcal{L}(M)$
($\because$ $M \underset{\text{simplify}}{\rightsquigarrow} \widetilde{M}$, $A^{\bullet}(M)\underset{\text{fact}}{\cong}A^{\bullet}(\widetilde{M})$)

Ⓘ Tropical modification, Deletion-Contraction
Tropical analogue of taking the graph of an algebraic function
$a\in E$ : fix
($M\setminus a$ : deletion by $a$
$\mathcal{L}(M\setminus a):=\{F\setminus\{a\} ; F\in\mathcal{L}(M)\}$
($M/a$ : contraction by $a$
$\mathcal{L}(M/a):=\{F\setminus\{a\} : F\in\mathcal{L}(M), a\in F\}$
- $M\setminus a$, $M/a$ : loopless matroids on $E\setminus\{a\}$ ← ($\because$ $\{a\}\in\mathcal{L}(M)$)
- $\Sigma_{M/a} \underset{\text{subfan}}{\subset} \Sigma_{M\setminus a}$

- $\bigoplus_{i\in E}\mathbb{Z}e_i \twoheadrightarrow \bigoplus_{i\in E\setminus\{a\}}\mathbb{Z}e_i \underset{\text{induce}}{\rightsquigarrow} N_E \twoheadrightarrow N_{E\setminus\{a\}} \underset{\text{induce}}{\rightsquigarrow} \Sigma_M \longrightarrow \Sigma_{M\setminus a}$

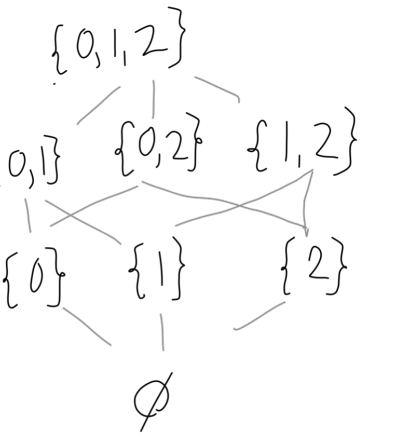
Example

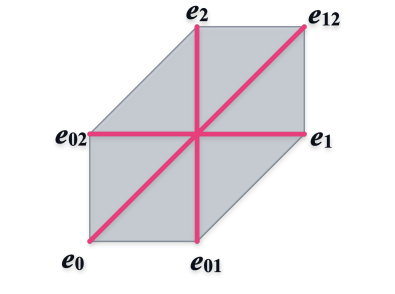




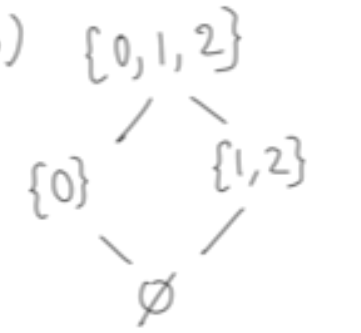
$\mathcal{L}(M)$

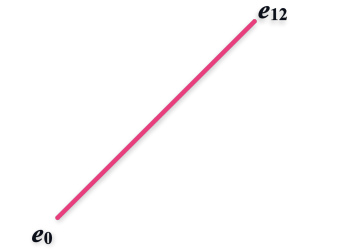
del by 3 →

$\mathcal{L}(M\setminus 3)$


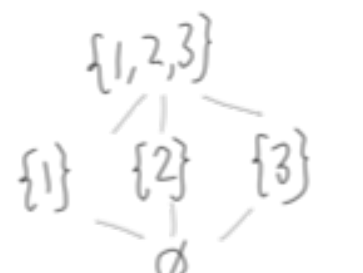


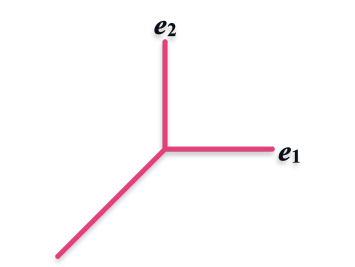
cont by 3 →

$\mathcal{L}(M/3)$




Del
cont by 0 →

$\mathcal{L}(M\setminus 0)=\mathcal{L}(M/0)$




$a\in E$ is called a coloop
$\overset{\text{def}}{\iff}$ $\mathrm{rank}(M)>\mathrm{rank}(M\setminus a)$
(i.e., $\dim\Sigma_M>\dim\Sigma_{M\setminus a}$)

Previous example
1, 2, 3 are non-coloop.
0 is coloop.

FACT
If $\forall a\in E$ is a coloop,
then $M=B_n$ (Boolean)
- $X(\Sigma_{B_n})$ is smooth projective,
  $\underset{\text{fact}}{\rightsquigarrow}$ $A^{\bullet}(B_n)\cong H^{2\bullet}(X(\Sigma_{B_n}))$
  $\rightsquigarrow$ PD of $A^{\bullet}(B_n)$ : OK
- We assume that $M\neq B_n$
- We fix $a\in E$ : non-coloop

def $S\subset E$
$\mathrm{cl}_M(S) := \bigcap\{F\in\mathcal{L}(M) : S\subset F\}$
closure of $S$ in $M$

Consider piecewise-linear map

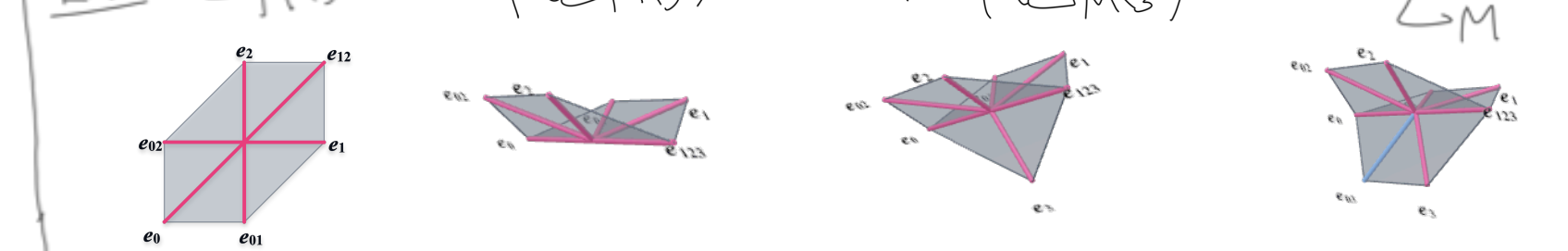
$$\begin{tikzcd}
\varphi : |\Sigma_{M\setminus a}| \arrow[r] & N_{E,\mathbb{R}} \\
e_G \arrow[u, phantom, "\cup" rotate=90] \arrow[r, mapsto] & e_{\mathrm{cl}_M(G)} \arrow[u, phantom, "\cup" rotate=90]
\end{tikzcd}$$

$(G\in\mathcal{L}(M\setminus a))$

$e_{\mathrm{cl}_M(G)} = \begin{cases} e_G & (G\in\mathcal{L}(M)) \\ e_G+e_a & (G\notin\mathcal{L}(M)) \end{cases}$

- $\varphi$ is a graph map of a piecewise-linear function on $\Sigma_{M\setminus a}$

Def
$TM_{\varphi}(\Sigma_{M\setminus a})$ ← graph of p.-lin. fct on $\Sigma_{M\setminus a}$
$:= \{\varphi(\sigma) ; \sigma\in\Sigma_{M\setminus a}\}\cup\{\varphi(\sigma)+\mathbb{R}_{\geq 0}e_a ; \sigma\in\Sigma_{M/a}\} \underset{\text{fan}}{\subset} N_{E,\mathbb{R}}$
called a tropical modification of $\Sigma_{M\setminus a}$ along $\Sigma_{M/a}$
- Bergman fans $\Sigma_M$ are tropical varieties
- $\varphi(\Sigma_{M\setminus a})$ (= left sum of $TM_{\varphi}(\Sigma_{M\setminus a})$) is not tropical
- $TM_{\varphi}(\Sigma_{M\setminus a})$ is the tropical closure of $\varphi(\Sigma_{M\setminus a})$
- (FACT, AP'20) $\pi : TM_{\varphi}(\Sigma_{M\setminus a})\to\Sigma_{M\setminus a}$ induces
  $\pi^{*} : A(M\setminus a) \xrightarrow{\sim} A(X(TM_{\varphi}(\Sigma_{M\setminus a})))$
- (FACT) $|TM_{\varphi}(\Sigma_{M\setminus a})| = |\Sigma_M|$

Exa. $\Sigma_{M\setminus 3} \underset{\text{graph}}{\rightsquigarrow} \varphi(\Sigma_{M\setminus 3}) \subset TM_{\varphi}(\Sigma_{M\setminus 3}) \xleftarrow{\text{star subdiv.}} \Sigma_M$


Ⓘ Blowups

FACT (Hsiao '25)
$TM_{\varphi}(\Sigma_{M\setminus a}) =: \Sigma^{(0)} \leftarrow \Sigma^{(1)} \leftarrow \cdots \leftarrow \Sigma^{(m)} = \Sigma_M$
Star subdivision at a 2-dim'l cone
(blowup at a torus inv. subspace of codim 2)

- Blowup center of $X(\Sigma^{(k)}) \longrightarrow X(\Sigma^{(k-1)})$
  $\cong X([\emptyset, F_k]_M)\times X([F_k\cup\{a\}, E]_M)$
  for some $F_k\in\mathcal{L}(M)$ with $\begin{cases} a\notin F_k, \\ F_k\cup\{a\}\in\mathcal{L}(M) \end{cases}$

Inductive hypothesis on $n=|E|$
$\rightsquigarrow$ $A^{\bullet}(X(TM_{\varphi}(\Sigma_{M\setminus a})))\cong A^{\bullet}(M\setminus a)$ : PD of dim $r$
$A^{\bullet}(\mathrm{Bl\ cent})\cong A([\emptyset,F_k]_M)\otimes A^{\bullet}([F_k\cup\{a\}, E]_M)$ : PD of dim $r-2$

Ⓘ Consequence of Keel's lemma.
In general, We consider

$$\begin{tikzcd}
\text{excep div} \arrow[r] & E \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\square"] & \widetilde{X} \arrow[d] & \text{Blowup of } X \text{ along } Z \arrow[l] \\
\text{closed subvar of codim} = c \arrow[r] & Z \arrow[r, hook, "i"'] & X & \text{smooth variety} \arrow[l]
\end{tikzcd}$$

if $\begin{cases} i^{*} : A^{\bullet}(X)\to A^{\bullet}(Z) \text{ is surjective} \\ [Z]\cdot A^{r-c}(X) = A^{r}(X) \\ A^{\bullet}(X) \text{ satisfies PD of dim } r \\ A^{\bullet}(Z) \text{ satisfies PD of dim } r-c \end{cases}$
then $A^{\bullet}(\widetilde{X})$ satisfies PD of dim $r$

$TM_{\varphi}(\Sigma_{M\setminus a}) = \Sigma^{(0)} \leftarrow \cdots \leftarrow \Sigma^{(m)} = \Sigma_M$
- $X(\Sigma^{(k-1)}) \leftarrow X(\Sigma^{(k)})$ : Blup at torus inv. subsp of codim 2
- $A(X(\Sigma^{(k-1)})) \twoheadrightarrow A\left(\begin{matrix}\text{Blow up center of} \\ X(\Sigma^{(k-1)}) \leftarrow X(\Sigma^{(k)})\end{matrix}\right)$
  ($\because$ these are toric variety and torus inv subspace)
- $A^{\bullet}(X(\Sigma^{(k-1)}))$ : PD of dim $r$
- $A^{\bullet}\left(\begin{matrix}\text{Blowup center of} \\ X(\Sigma^{(k-1)}) \leftarrow X(\Sigma^{(k)})\end{matrix}\right)$ : PD of dim $r-2$

$\rightsquigarrow$ $A^{\bullet}(\Sigma^{(k)})$ : PD of dim $r$
$\therefore$ $A^{\bullet}(M)$ satisfies PD of dim $r$ //

App
Keel's lemma (1972

$$\begin{tikzcd}
\text{excep. div.} \arrow[r] & E \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\square"] & \widetilde{X} \arrow[d, "\pi^{*}"] & \text{Blowup of } X \text{ along } Z \arrow[l] \\
\text{closed subvar of codim} = c \arrow[r, bend left] & Z \arrow[r, hook, "i"'] & X & \text{smooth variety} \arrow[l, bend right]
\end{tikzcd}$$

If $i^{*} : A^{\bullet}(X)\to A^{\bullet}(Z)$ is surjective ($\frac{A^{\bullet}(X)}{\mathrm{Ker}\, i^{*}}\cong A^{\bullet}(Z)$),
then $\widetilde{\pi}^{*} : A^{\bullet}(X)[T] \longrightarrow A^{\bullet}(\widetilde{X})$
$T \longmapsto -[E]$
induces an isom. $\dfrac{A^{\bullet}(X)[T]}{(P(T),\, T\cdot\mathrm{Ker}\, i^{*})} \xrightarrow{\sim} A^{\bullet}(\widetilde{X})$
monic polynomial of degree $c$, costant term $=[Z]$

(Chern poly : $i^{*}P(T) = T^{\ell}+c_1(N_Z X)T^{\ell-1}+\cdots+c_{\ell}(N_Z X)$)

obs
$\forall q\in\mathbb{Z}_{\geq 0}$
$A^{q}(\widetilde{X}) \cong \dfrac{A^{q}(X)\oplus T\cdot A^{q-1}(X)\oplus\cdots\oplus T^{c-1}\cdot A^{q-c+1}(X)}{(T\cdot\mathrm{Ker}(i^{*}))_{q}}$
$\cong A^{q}(X)\oplus T\cdot A^{q-1}(Z)\oplus\cdots\oplus T^{c-1}\cdot A^{q-c+1}(Z)$

Assume $\begin{cases} A^{\bullet}(X) \text{ satisfies PD of dim } r \\ A^{\bullet}(Z) \text{ satisfies PD of dim } r-c \end{cases}$

Aim : $A^{\bullet}(\widetilde{X})$ satisfies PD of dim $r$

- $q>r \Longrightarrow \begin{pmatrix} A^{q}(X)=0 \\ A^{q-1}(Z)=\cdots=A^{q-c+1}(Z)=0 \end{pmatrix}$ ← $q-c+1>r-c$
  $\rightsquigarrow A^{q}(\widetilde{X})=0$ by
- $q=r \Longrightarrow A^{r}(\widetilde{X})\cong A^{r}(X)\cong\mathbb{Z}$

Assume that $q<r$
$A^{q}(\widetilde{X}), A^{r-q}(\widetilde{X})$ are finitely generated free abelian group by PD of $A(X)$ and $A(Z)$ and

Consider the matrix of
$A^{q}(\widetilde{X})\times A^{r-q}(\widetilde{X}) \longrightarrow A^{r}(\widetilde{X})\cong\mathbb{Z}$

$$\begin{array}{c|cccc}
 & A^{r-q}(X) & A^{r-q-1}(Z) & \cdots & A^{r-q-c+1}(Z) \\
\hline
A^{q}(X) & P_0 & 0 & \cdots & 0 \\
A^{q-1}(Z) & 0 & & & \\
\vdots & \vdots & & & \\
A^{q-c+1}(Z) & 0 & & &
\end{array}$$

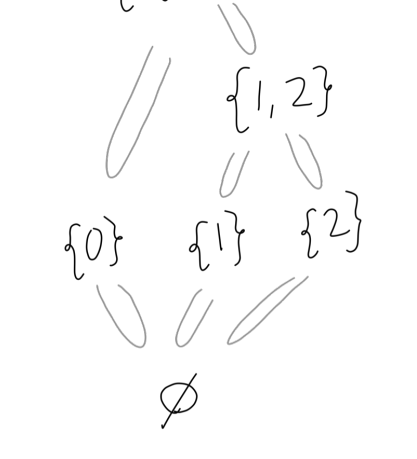
$P(T)=T^{c}+\cdots+ \,T\, \boxed{+[Z]}$
$[Z] = 0$

$\begin{cases} T\cdot\alpha \in T\cdot A^{q-1}(X) \\ T^{c-1}\cdot\beta \in T^{c-1}A^{r-q-c+1}(X) \end{cases}$

$(T\cdot\alpha)(T^{c-1}\cdot\beta) = \boxed{T^{c}}\alpha\beta \in A^{r-c}(X)$
$\equiv$ mod $T\cdot\mathrm{Ker}\, i^{*}$

$i^{*} : A^{r-c}(X) \longrightarrow A^{r-c}(Z)$
$\alpha\beta \longmapsto i^{*}(\alpha\beta)$

$i_{*}(i^{*}(\alpha\beta)) = i_{*}(i^{*}(\alpha\beta)\cdot 1)$ ($i_{*}1 = [Z]$)
$\underset{\text{proj. formula}}{=} \alpha\beta\cdot[Z]$



$\begin{cases} F=\{1\} \\ 0\in E\setminus F=\{0,2\} \end{cases}$